Continuous Random Variables

Section 1: Probability Density Functions (PDF)

1 A continuous random variable X has PDF $f(x) = kx(2 - x)$ for $0 \le x \le 2$, and 0 otherwise. Find the value of k.

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{8}$

d. 1

2 For the random variable in the previous question ($k = 3/4$), calculate $P(0 \le X \le 1)$.

a. $\frac{1}{2}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{5}{8}$

3 What is the probability that a continuous random variable takes any single exact value, such as $P(X = c)$?

a. 0

b. 1

c. $f(c)$

d. $f'(c)$

Section 2: Cumulative Distribution Function $F(x)$

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4 Given PDF $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, find the cumulative distribution function $F(x)$ for $0 \le x \le 2$.

a. $F(x) = \frac{x^3}{8}$

b. $F(x) = \frac{3x^3}{8}$

c. $F(x) = \frac{x^2}{4}$

d. $F(x) = \frac{x^3}{2}$

5 For the distribution in the previous question, find the median m.

a. $m = \sqrt[3]{4} \approx 1.587$

b. $m = 1.0$

c. $m = \sqrt{2} \approx 1.414$

d. $m = 1.5$

6 Given $F(x) = \frac{x^2 - 1}{8}$ for $1 \le x \le 3$, find the PDF $f(x)$ on this interval.

a. $f(x) = \frac{x}{4}$

b. $f(x) = \frac{x}{8}$

c. $f(x) = \frac{x^2}{8}$

d. $f(x) = \frac{2x - 1}{8}$

Section 3: Mean, Variance & Medians

7 Find the expected value $E(X)$ for the PDF $f(x) = 2x$ on $[0, 1]$.

a. $\frac{2}{3}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

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8 For the distribution $f(x) = 2x$ on $[0, 1]$, calculate the variance $\text{Var}(X)$.

a. $\frac{1}{18}$

b. $\frac{1}{9}$

c. $\frac{1}{12}$

d. $\frac{2}{9}$

9 Find the mode of the continuous distribution with PDF $f(x) = 12x^2(1 - x)$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

Section 4: Piecewise Distributions & Examination Standards

10 A continuous random variable has PDF $f(x) = cx$ for $0 \le x < 2$ and $f(x) = c(4 - x)$ for $2 \le x \le 4$. Find c.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{8}$

d. 1

11 For the symmetric triangular distribution in the previous question, what is $E(X)$?

a. 2

b. 1

c. 3

d. 2.5

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12 Find the lower quartile Q_1 for the triangular distribution in the previous question ($c = 1/4$).

a. $\sqrt{2} \approx 1.414$

b. 1.0

c. 1.5

d. $\sqrt{3} \approx 1.732$

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